YouTube Trending Video Analytics
PROJECT REPORT

Introduction

YouTube's trending list reflects what audiences are watching and engaging with at any given moment, but the patterns behind it — which categories dominate, how sentiment relates to views, and how activity differs across regions — are not obvious without analysis. This project examines trending video data from five countries (United States, United Kingdom, India, Japan, and Germany) to uncover these patterns and present them through SQL analysis, Python-based exploration, and an interactive Tableau dashboard.

Abstract

This project analyzes 178,580 trending video records spanning November 2017 to June 2018, sourced from Kaggle's YouTube trending video datasets. After cleaning and merging country-level data, ten SQL queries were used to explore category trends, view and engagement patterns, and cross-country video reach. Title sentiment was classified using VADER (with tags scored separately as a supplementary check), and the resulting patterns — along with category and time-based trends — were visualized in a Tableau dashboard. The analysis found that negative-sentiment titles draw nearly double the average views of neutral titles, and that a small set of major movie trailers and global music releases account for the videos that trend simultaneously across all five countries.

Tools Used

- Python (Pandas, Matplotlib, Seaborn) — data cleaning, merging, and visualization
- SQL (DB Browser for SQLite) — querying and aggregating trending video data
- VADER (NLTK) — sentiment analysis on video titles
- Tableau Public — interactive dashboard design and publishing

Steps Involved in Building the Project

- Loaded and cleaned trending video CSVs for five countries in a Jupyter notebook
- Merged country-level data and exported it to SQLite for SQL-based analysis
- Wrote and documented ten SQL queries covering views, engagement, category trends, and cross-country reach
- Applied VADER sentiment analysis to video titles (with tags analyzed separately as a supplementary check) and grouped title-based results into Positive, Neutral, and Negative categories
- Built exploratory visualizations in Python using Matplotlib and Seaborn
- Designed an interactive Tableau dashboard covering category, sentiment, and time-based trends
- Published the dashboard to Tableau Public and compiled findings into a final report

Conclusion

This project shows that trending video performance is shaped by more than category alone — title sentiment and cross-country reach play a measurable role in how far a video travels. Negative or provocative titles consistently outperform neutral ones in average views, while genuine global virality is rare and concentrated in major entertainment releases rather than everyday content. Combining SQL, Python, and Tableau into a single workflow made it possible to move from raw, multi-country data to a clear, presentable set of insights.

Dashboard link: View on Tableau Public